UNDERSTANDING AND DETECTING FACTUAL HALLUCINATIONS IN RAG THROUGH RESPONSE PATTERN ANALYSIS

PRESENTED BY
Gentili Camilla (888517), Princic Elisa (886476), Torre Andrea (886051)

ARTIFICIAL INTELLIGENCE
A.A. 2025 - 2026

PROBLEM

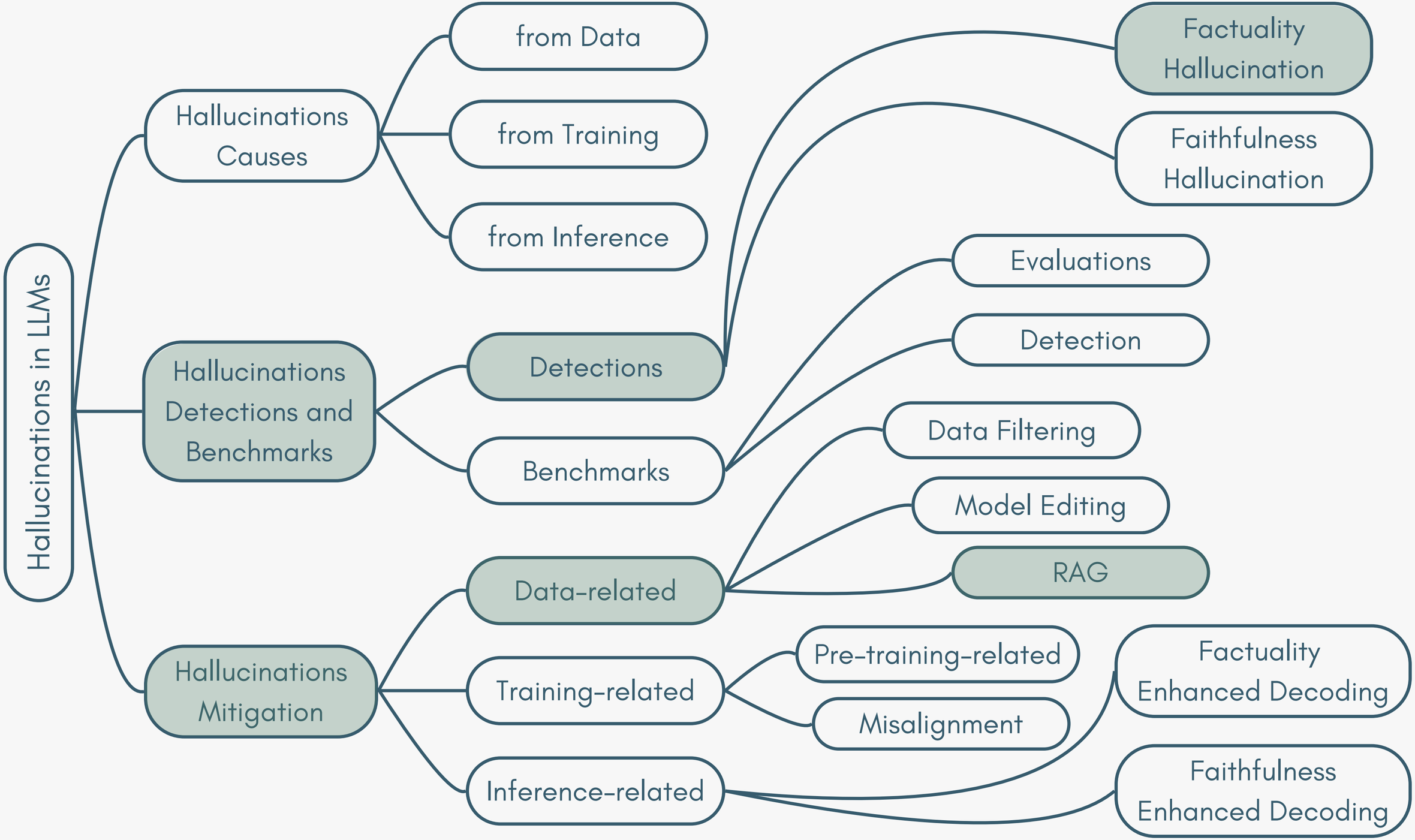
LLMs generate plausible, authoritative explanations, but they may introduce unsupported facts. Hallucinations may be hard to spot because the writing is internally coherent.

GOAL

We want to analyze hallucinations through response patterns. Beyond accuracy, we study how hallucinations happen and what they look like.

METHODOLOGY

RAG: Question → Retrieval → Context → LLM → Answer


Hallucinations in LLMs
Hallucinations Causes
from Data
from Training
from Inference
Hallucinations Detections and Benchmarks
Detections
Benchmarks
Factuality Hallucination
Faithfulness Hallucination
Evaluations
Detection
Data Filtering
Model Editing
RAG
Hallucinations Mitigation
Data-related
Training-related
Pre-training-related
Misalignment
Inference-related
Factuality Enhanced Decoding
Faithfulness Enhanced Decoding

Why Factual Hallucinations matter

In a closed domain, the key risk is not fluency but **incorrect**, **confident** factual statements. These errors can be subtle: the answer sounds plausible while adding or altering details not supported by evidence.

→ Measuring overall accuracy alone hides how the model fails; we want recurring patterns of factual mistakes.

A **factual hallucination** is a claim in the response that is not supported by the retrieved evidence (RAG setting), or not supported by the ground-truth reference (when available).

Example

Q: What is the capital of Italy?

A: Milan is the capital of Italy.

Strage di Bologna Dataset

The dataset contains **fact-oriented questions** about the Strage di Bologna in a **closed domain** setting.
The source material consists of four documents provided by the Department of Computer Science, distributed as text files:

- sentenza_processo_d_appello_18_07_1990.txt
- motivi_ricorso_cassazione_27_05_1991.txt
- sentenza_cassazione_12_02_1992.txt
- sentenza_def_cassaz_23_11_1995.txt

These documents form the closed knowledge base on which the RAG pipeline is built: they are used to retrieve the contexts used to generate responses and to assess their accuracy and fidelity to the sources.

Additionally, the 20 questions (five per txt) and the ground truth were provided.

Pipeline Overview

1. **Domain Setup**: Collect a dense, topic-focused set of documents to ensure a controlled, low-ambiguity environment.
2. **Preprocessing & Chunking:** Clean documents and split them into overlapping chunks (fixed chunk size + overlap) to preserve local context.
3. **Indexing**: Create embeddings for chunks and build a vector index (optionally combined with lexical retrieval) for efficient retrieval.
4. **Question Set / Prompts**: Prepare a set of evaluation questions and define prompting rules (e.g., “answer only from provided context; admit uncertainty if missing”).

5. **Retrieval (Top-k Context)**: For each question, retrieve the top-k most relevant chunks (dense or hybrid retrieval).
6. **Generation** of RAG LLM*: answer conditioned on retrieved context (external knowledge injected) .
7. **Logging & Traceability**: Store for each query: retrieved chunks, final prompt, model response, and metadata (e.g., source documents).
8. Evaluation (Automatic + Manual):
    - **Automatic** hallucination-related metrics (e.g., RAGAS; faithfulness vs. context, agreement vs. reference).
    - **Manual** review to validate edge cases and refine taxonomy.

*LLM used for the project : *llama-3.3-70b-versatile*

Retrieval-Augmented Generation (RAG) Implementation

PURPOSE: RAG encourages evidence-based answers. Every answer can be linked to retrieved passages, making factual errors easier to identify, explain, and categorize.

Index and contexts were build for each txt source independently.

Each source document is split into overlapping chunks to preserve continuity across boundaries.
For every document, a dedicated index folder is created storing:

- Chunk texts + metadata
- Embedding matrix
- Sparse lexical index

NB: One of the four provided source is empty → empty contexts → index not created (no retrieval possible)

→ **Engine (3 strategies per index and retrieval)**

- **Dense** search: normalized embeddings + FAISS (inner product uses cosine similarity)
- **Sparse** search: BM25 over simple tokenization (keyword matching)
- **Hybrid** search: dense + sparse scores normalized and summed to rank chunks

→ **RAG execution**

Inputs from Excel: question + source

1. Load available indices from the base directory
2. Select the source-specific index
3. Retrieve Top-k contexts
4. Build the prompt (we tested two prompt variants)
5. Generate the answer and log answer + Top-k retrieved contexts

Theoretical intuition

RAG estimates chunk relevance via:

- **Semantic** relevance (dense): cosine similarity between query and chunk embeddings
- **Lexical** relevance (sparse): rewards chunks containing key terms from the query

Evaluation: RAGAS

RAGAS is an open source framework for testing and evaluating LLM applications. It combines LLM-driven metrics with systematic experimentation to create a continuous improvement loop.

It measures:

- Correctness vs. Ground Truth (Reference-based)
  - **False Positives** (FP): statements present in the answer but not supported by the ground-truth reference
  - **False Negatives** (FN): statements present in the ground truth but missing from the answer

  → Captures what the model adds incorrectly vs what it fails to cover.

FN are **not considered** in the analyses because we want to evaluate the hallucinations, not the **incompletness** of the RAG answers.

- Faithfulness vs. Retrieved Context (Evidence-based)
  - Checks whether answer statements are supported by the retrieved passages.
  - Low faithfulness indicates unsupported claims (hallucinations in the RAG setting).

We used two prompt variants:

Prompt A — "**Context-only**": Answer using only the information in the retrieved context.

Prompt B — "**Context-assisted**": Answer with the help of the retrieved context.

Table 1: Prompt Context-only

Domanda	Precision	Recall	Faithfulness
1	0.60	0.60	1.00
2	0.50	0.50	1.00
3	0.50	0.33	1.00
4	1.00	1.00	1.00
5	0.67	0.50	0.50
6	0.67	0.67	1.33
7	0.67	0.40	0.00
8	0.67	0.67	0.67
9	0.67	0.33	1.00
10	1.00	0.50	1.00
11	1.00	0.66	1.00
12	1.00	0.66	1.00
13	1.00	0.50	0.67
14	0.50	0.50	0.50
15	0.00	0.00	0.50
mean	0.63	0.53	0.79
median	0.67	0.50	1.00

Table 2: Prompt Context-assisted

Domanda	Precision	Recall	Faithfulness
1	0.67	0.80	1.00
2	0.00	0.00	1.00
3	0.00	0.00	1.00
4	0.67	1.00	0.67
5	0.00	0.00	0.00
6	0.25	0.25	0.25
7	0.25	0.50	0.75
8	0.67	0.67	1.00
9	0.00	0.00	1.00
10	1.00	1.00	1.00
11	1.00	0.67	1.00
12	0.50	0.50	1.00
13	0.25	0.50	1.00
14	0.00	0.00	0.50
15	0.00	0.00	0.50
mean	0.46	0.49	0.78
median	0.50	0.50	1.00

- Precision and Recall are computed against the Ground Truth (GT) while faithfulness against context

A statement can be factually correct but absent from the GT, and it will still be treated as an addition (FP) or reduce agreement.

EMPTY TXT

Consequence:

- 0 chunks produced during preprocessing
- Index is not created
- Retrieval returns empty contexts for queries targeting that source

Model behavior Observed: RAG with both prompt refused to answer as the no context was available for the system.

Another example of modification that could be applied to RAGAS:

"**faith_verdicts**":
{
"**statement**": "Non posso rispondere.",
"**verdict**": 0,
"**reason**": "Il contesto è \nan\ e non fornisce informazioni sufficienti per confermare o negare la dichiarazione."
}

Human Evaluation

GOAL: Validate and interpret RAGAS False Positives (FPs) through manual inspection, to distinguish true hallucinations from evaluation artifacts.

Procedure

1. Inspect the question, the model answer, the retrieved context, and the ground truth.
2. Verify whether the FP claim is:
    - Unsupported by both context and reference (likely hallucination)
    - Reasonable but penalized due to missing/incomplete ground truth
    - Caused by “no-answer” behavior when evidence is insufficient
3. Assign each FP to a manual category and attach notes.

We produced a set of FP cases labeled into 3 categories, used for qualitative pattern analysis and for interpreting aggregate metrics.

No Answer Retrieved / Missing Evidence

The RAG pipeline fails to retrieve passages that contain a direct answer, especially for questions asking for motivations / explanations rather than explicit facts


```
{
  "statement": "Le informazioni fornite non menzionano il procuratore Squadrini o
                le motivazioni relative a Musumeci e Belmonte.",
  "verdict": 0,
  "reason":   "Il contesto menziona espressamente il procuratore Squadrini e le
               motivazioni relative a Musumeci e Belmonte, ovvero l'esclusione
               dell'aggravante prevista dall'art. 1 della Legge 6 febbraio 1980 n. 15
               per il delitto di calunnia."
}
```

Ground Truth Missing Detail (Evaluation Artifact)

The model provides a detail requested by the question, but that detail is not explicitly stated in the ground truth reference.


```
{
    "statement": "La Corte Suprema di Cassazione a Sezioni Unite Penali ha annullato la
                  sentenza impugnata nei confronti di Pazienza Francesco.",
    "verdict": 0,
    "reason":   "La Corte di Cassazione ha annullato la sentenza impugnata, ma non è
                  specificato che si tratti della Corte Suprema di Cassazione a Sezioni Unite
                  Penali."
}
```

Complete Fabrication (True Hallucination)

The answer introduces a claim that is not supported by retrieved context and not present in ground truth.


```
{
  "statement": "La formula utilizzata per assolvere il Rinani dal delitto di strage nel 1990 è
                l'assoluzione con formula piena.",
  "verdict": 0,
  "reason": "Il contesto afferma che Il Rinani veniva assolto con la formula del dubbio, non con
             formula piena."
}
```

Summary table

Prompt	No Answer	GT Missing Detail	True Hallucination
Context-only	5	5	6
Context-assisted	2	11	10

Questions on empty txt were excluded

RAGAS LIMITS

1. Comparing the RAG with different prompts, LLM judge in RAGAS had different verdicts on same answer
2. Misleading motivations
3. Faithfulness sometimes lacks coherence

{
"statement": "Secondo la regola di specialità dell'estradizione, Gelli non può essere processato per reati diversi da quelli per cui è stato estradato."

context-only

"verdict": 1,
"reason": "Il contesto non menziona la regola di specialità dell'estradizione, ma è un concetto giuridico noto che si applica in questo caso."

context-assisted

"verdict": 0,
"reason": " Il contesto non menziona la regola di specialità dell'estradizione, ma solo che Gelli era stato estradato per reati diversi.
}

WORKFLOW LIMITS

1. As we initially thought, we did not find **position** informative in our case:
    a. Top-k-context are ordered
    b. Txt is influenced by natural order
2. Quantitative results are not reliable
3. **Dataset**
    a. Complex terminology
    b. Temporal dependencies
    c. Ambiguities:
        i. Negation scope
        ii. Referential
        iii. Anaphora
    d. Syntactics (use of subordinates)

CONCLUSIONS

1. The LLM sometimes produced **inconsistent interpretations** of the same underlying result, suggesting sensitivity to prompting and/or retrieved evidence
2. The dataset provides **limited reference points.** Better understanding of speaker and temporal dependencies in the construction of RAG and context. Map roles of subjects, etc
3. Quantitative outputs (especially **metrics and numerical claims**) require additional verification: we observed cases where numbers look suspicious **or internally inconsistent**, motivating targeted checks and stricter validation
4. Struggles when asked ‘i motivi’ in the questions
5. More abstention is needed

REFERENCES

- Lei Huang et al. (2024): A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions https://arxiv.org/pdf/2311.05232

- RAGAS: https://docs.ragas.io/en/stable/

- Sewong Min et al. (2023): FACTSCORE: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation https://arxiv.org/pdf/2305.14251

- Ceccarelli et al. (2025): Knowledge-Grounded Detection of Factual Hallucinations in Large Language Models https://clic2025.unica.it/wp-content/uploads/2025/09/22_main_long.pdf

THANK YOU !